Goal 1: To understand the relationships between relations, implicit differentiation, slopes and tangent lines.

Goal 2: To be able to differentiate implicitly and use our prior knowledge and skills to do our FIRST AP FREE RESPONSE QUESTION ..... YAY!!!!

2. Consider the curve given by $y^2 = 2+xy$.

(a) Show that $\frac{dy}{dx}=\frac{y}{2y-x}$

(b) Find all points (x, y) on the curve where the line tangent to the curve has a slope $\frac{1}{2}$.

(c) Show that there are no points (x, y) on the curve where the line tangent to the curve is horizontal.

6. Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(a) Show that $\frac{dy}{dx} = \frac{-(x+1)}{2(y^3+1)}$.

(b) Write an equation for the line tangent to the curve at the point $(-2, 1)$.

(c) Find the coordinates of the two points on the curve where the line tangent to the curve is vertical.

(d) Is it possible for this curve to have a horizontal tangent at points where it intersects the x-axis? Explain your reasoning.

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Question 6

Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(a) Show that $\dfrac{dy}{dx} = \dfrac{-(x+1)}{2\left(y^3+1\right)}$.

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Question 6

Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(b) Write an equation for the line tangent to the curve at the point $(-2, 1)$.

$$\frac{dy}{dx} = \frac{-(x+1)}{2\left(y^3+1\right)}$$

Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(c) Find the coordinates of the two points on the curve where the line tangent to the curve is vertical.

$$\frac{dy}{dx} = \frac{-(x+1)}{2\left(y^3+1\right)}$$

Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(d) Is it possible for this curve to have a horizontal tangent at points where it intersects the x-axis? Explain your reasoning.

$$\frac{dy}{dx} = \frac{-(x+1)}{2\left(y^3+1\right)}$$

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6. Consider the curve given by the equation $y^3 - xy = 2$. It can be shown that $\frac{dy}{dx} = \frac{y}{3y^2 - x}$.

   (a) Write an equation for the line tangent to the curve at the point $(-1, 1)$.

   (b) Find the coordinates of all points on the curve at which the line tangent to the curve at that point is vertical.

   (c) Evaluate $\frac{d^2y}{dx^2}$ at the point on the curve where $x = -1$ and $y = 1$.